EXECUTIVE BRIEF · VERSION 1.0 · OCTOBER 2026

NeoStay

The short version

Forecast how long a very small baby will stay, and staff the neonatal unit for the acuity actually present — from the unit's own data and its own policy. Four pages: what it does, what we want to prove, what we need, and the decisions we are asking for.

Waseem Altaf, MD

Project and Clinical Lead · Neonatologist

Abdul Razak, PhD

Technical Lead, Lead Developer and Applied AI/Data Science Lead

DOCUMENT
Executive brief

AUDIENCE
Hospital executive and service managers

DETAIL
Full dossier and data request available

STATUS
Decision-support prototype — not a medical device

01 The case in one page

Problem, answer, and where the project stands.

The problem. Neonatal units plan two things under uncertainty: how long each baby will stay, and how many skilled nurses each shift needs. Both are usually judged from experience, and neither leaves evidence behind when an establishment is questioned.

The answer. NeoStay turns a decade of the unit's own recorded outcomes into an expected length of stay with a realistic range, and turns the unit's own acuity policy into nurses per shift, the required skill mix, and a 24-hour roster that names any cot left without a qualified nurse. It runs in a web browser, behind sign-in, with no patient-level data.

712
VLBW INFANTS BEHIND THE FIGURES

108
ACUITY FINDINGS ENCODED

<5 min
TO CLASSIFY A FULL UNIT

111
AUTOMATED TESTS PASSING

What exists today

Capability	Status
Length of stay, discharge outcome, survival, with confidence	**Working** — interpolated from the unit's recorded statistics
Acuity classification from the unit's own tool	**Working** — 11 body systems, 64 level-of-care criteria
Nurses per shift, skill mix, 24-hour roster, uncovered cots	**Working** — rules-based, policy-driven
Security: named accounts, encrypted published copy, audit-ready deployment	**Working**
Fitted statistical or machine-learning models	**Not yet** — requires hospital data (Section 3)
Prospective clinical validation	**Not yet** — planned, Phase 2

Stated plainly. NeoStay is a decision-support prototype. It is not a medical device, it has not been clinically validated, and it never replaces clinical judgment. Every recommendation is advisory and requires a human decision.

02 What we want to prove, and how

The analytical plan in brief. Full detail in the dossier.

TABLE 1 — FOUR QUESTIONS, FOUR METHODS, ONE WAY TO JUDGE THEM

Question	Models to be tested	Judged by
How long will this baby stay?	Survival models with competing risks (home, transfer, death); gradient boosting; quantile and Bayesian hierarchical variants	Error in days and calibration, against today's method
What care level will this baby need next shift?	Multi-state and ordinal models; cost-weighted gradient boosting	Accuracy, with under-calling penalised
What will the unit need over the next fortnight?	Seasonal time-series and state-space models; queueing approximation; simulation	Error in nurse-hours, and whether the stated range holds
Who should take which cot tonight?	Constraint-based optimisation, with today's rule as fallback	Uncovered cots and planner acceptance

The rule we have set ourselves. A new model replaces the current method only if it is better on data it has never seen, stays calibrated, can be explained to the clinical lead, and runs fast enough to use at handover. Otherwise the simpler method stays. Everything is reported to the TRIPOD+AI standard, with performance published per birth-weight band and per branch.

Rollout in three phases

PHASE 1

Pilot

Configure one unit's grades and ratios, run in parallel with the current method for four weeks, measure task time and acceptance.

PHASE 2

Evidence

Historical data under governance; build and validate models; silent prospective run; published validation report.

PHASE 3

Group

Branch-by-branch rollout by configuration, roster optimisation, dashboards, annual revalidation.

03 What we need, and the decisions we are asking for

Short list. The data request is set out field by field in the companion document.

TABLE 2 — WHAT THE HOSPITAL PROVIDES

Item	Why	When
Historical admissions data, pseudonymised	To fit and validate the length-of-stay and acuity models; without it NeoStay stays deterministic	Phase 2 start
Census and staffing history	To forecast demand and compare recommended with actual staffing	Phase 2 start
Local nursing policy: grades, ratios, supervision rules	So the tool reflects this hospital, not the one it was written in	Phase 1
A pilot unit and two champions	Parallel running and honest feedback	Phase 1
Information-governance route: named data owner, DPIA, sharing agreement	Legal basis for any patient-level extract	Phase 1
Hosting: one small server or container platform, HTTPS, single sign-on if required	To run the service inside the hospital's boundary	Phase 1

Decisions requested

01 **Approve the pilot** in one neonatal unit, with parallel running for four weeks.

02 **Name a data owner** and start the governance route for a pseudonymised historical extract.

03 **Confirm the nursing policy** to configure: grade ladder, ratios, supervision rules.

04 **Choose the hosting model**: hospital server, hospital cloud tenancy, or managed container platform.

05 **Agree the success measures** in Table 1 of the dossier, so adoption is decided on evidence.

What the hospital gets, whatever the models show. Even before any model is fitted, the unit gains a consistent way to classify acuity, a defensible figure for nurses per shift and skill mix, a record of shifts where cover was short, and a shared reference for its own policy. The modelling work adds accuracy and foresight on top of that.

NeoStay — Executive Brief, version 1.0. Project and Clinical Lead: Waseem Altaf, MD, Neonatologist. Technical Lead, Lead Developer and Applied AI/Data Science Lead: Abdul Razak, PhD. Decision support only — not for clinical use. © 2026 Waseem

Altaf, MD. Source: `docs/architecture/neostay-executive-brief.html`.